1. Kuhn-Tucker

Utility maximization:

$$\mathcal{L} := U(\mathbf{x}) + \lambda[m - \mathbf{p}\cdot\mathbf{x}].$$

- $\partial\mathcal{L}/\partial x_i \le 0$, $x_i \ge 0$, with C.S.
- $\partial\mathcal{L}/\partial\lambda \ge 0$, $\lambda \ge 0$, with C.S.

Cost minimization:

$$\mathcal{L} := \boldsymbol{\omega}\cdot\mathbf{x} + \lambda[y - f(\mathbf{x})].$$

- $\partial\mathcal{L}/\partial x_i \ge 0$, $x_i \ge 0$, with C.S.
- $\partial\mathcal{L}/\partial\lambda \le 0$, $\lambda \ge 0$, with C.S.

2. Envelope Theorem

Theorem 2.1. *Consider $V(t) := \sup_{x\in X} f(x,t)$ and $X^*(t) = \{x \in X : f(x,t) = V(t)\}$. We have $V'(t) = f_t(x,t)$ for each $x \in X^*(t)$ if both sides exist.*

3. Consumer Theory

Proposition 3.1 (Slutsky)**.**

- $\frac{\partial x^m}{\partial p_x} = \frac{\partial x^h}{\partial p_x} - \frac{\partial x^m}{\partial m}\cdot x^m$.
- $\epsilon^m_{ij} = \epsilon^h_{ij} - \eta_i s_j$.
- $s_i\epsilon^m_{ij} = s_j\epsilon^m_{ji} + s_i s_j(\eta_j - \eta_i)$.

Proposition 3.2.

- *Engel aggregation: $\sum s_i\eta_i = 1$. It will never be the case that all goods are inferior, all goods are necessities, or all goods are luxuries.*
- *Cournot aggregation: $\sum_i \epsilon_{ij}s_i = -s_j$. A Giffen good has many complements; the budget share of good j is small if it has many substitutes, and large if it has many complements, or when their respective budget shares are large; if the number of substitutes for a good decrease, the demand for that good becomes more inelastic.*
- *Homogeneity: $\sum_i \epsilon_{ji} = -\eta_j$. A Giffen good is the substitute of many goods.*

4. Production Technology

Assumption 4.1. The production function $f : \mathbb{R}^m_+ \to \mathbb{R}_+$ is continuous, strictly increasing, and strictly quasiconcave on $\mathbb{R}^m_+$ and $f(\mathbf{0}) = 0$.

- Strict quasiconcavity gives strictly convex upper contours. It can be thought of as there being some complementarities in the inputs.
- These assumptions guarantees that the firm's cost minimization problem is well-defined and has a unique solution. (Continuity guarantees the existence of a solution to the cost-minimization problem when $\omega \gg 0$; strict quasi-concavity guarantees that the solution is unique.)

Definition 4.2. The **marginal rate of technical substitution (MRTS)** between inputs i and j is $\mathrm{MRTS}_{ij}(\mathbf{x}) := \mathrm{MP}_i(\mathbf{x})/\mathrm{MP}_j(\mathbf{x})$.

Proposition 4.3. *Under Assumption 4.1 and when $n = 2$,*

(i) *The slope of isoquant (at a point $\mathbf{x}$) is given by the MRTS.*
(ii) *Isoquant are always downward sloping.*
(iii) *We have diminishing* MRTS.

Definition 4.4. For a production function $f(\mathbf{x})$, the **elasticity of Substitution between inputs i and j** (at point $\mathbf{x}$) is defined as

$$\sigma_{ij}(\mathbf{x}) := -\frac{\mathrm{d}\log(x_i/x_j)}{\mathrm{d}\log(\mathrm{MP}_i/\mathrm{MP}_j)} = \frac{\mathrm{d}\log(x_j/x_i)}{\mathrm{d}\log(\mathrm{MP}_i/\mathrm{MP}_j)}.$$

Remark 4.5. A larger σ_{ij} means it is easier to substitute i for j. To see this, consider the CES production function. As $\sigma \to \infty$, $f(\mathbf{x}) \to \alpha x_1 + (1-\alpha)x_2$; as $\sigma \to 0$, $f(\mathbf{x}) \to \min\{x_1, x_2\}$.

5. Cost Minimization

Assuming interior solution, we have

$$\forall i, j : \frac{\omega_i}{\omega_j} = \frac{\mathrm{MP}_i}{\mathrm{MP}_j},$$

5.1. Comparative Statics.

The substitution matrix is

$$\boldsymbol{\sigma}^*(\boldsymbol{\omega}, y) := \left[\frac{\partial x_i(\boldsymbol{\omega}, y)}{\partial \omega_j}\right].$$

Proposition 5.1. *If the production function is CRS [IRS, DRS], then its average cost function is constant [decreasing, increasing].*

Proposition 5.2 (Cost Function)**.** *$c(\boldsymbol{\omega}, y)$ is*

- *Zero when $y = 0$.*
- *Continuous on its domain.*
- *If $\boldsymbol{\omega} \gg 0$, strictly increasing in y; thus marginal cost of production is always positive when there are no free inputs.*
- *Increasing, concave, and homogeneous of degree one in $\boldsymbol{\omega}$.*
- *(Shephard's Lemma) If f is strictly quasiconcave and $\boldsymbol{\omega} \gg 0$, then $\partial c(\boldsymbol{\omega}, y)/\partial\omega_i = x_i(\boldsymbol{\omega}, y)$.*

Proposition 5.3 (Conditional Input Demand Function)**.**

- *$\mathbf{x}(\boldsymbol{\omega}, y)$ is homogeneous of degree zero in $\boldsymbol{\omega}$.*
- *Assuming differentiability, $\partial x_i(\boldsymbol{\omega}, y)/\partial y \ge 0$; conditional input demand is never inferior.*
- *The substitution matrix is symmetric and negative semidefinite. In particular, the negative semidefiniteness property implies that $\partial x_i/\partial\omega_i \le 0$ for each i, law of demand.*

5.2. Short-run Vs. Long-run Cost Minimization.

Proposition 5.4. *For every $(\boldsymbol{\omega}, y)$, we have $c^{SR}(\boldsymbol{\omega}, y; \overline{\mathbf{x}}) \ge c(\boldsymbol{\omega}, y)$, with equality when $\overline{\mathbf{x}} = \mathbf{x}(\boldsymbol{\omega}, y)$, the long run conditional input demand. In particular:*

- *The long-run cost curve is the lower envelope of the family of short-run cost curves.*
- *We have the identity $c^{SR}(\boldsymbol{\omega}, y; \mathbf{x}(\boldsymbol{\omega}, y)) = c(\boldsymbol{\omega}, y)$.*

Corollary 5.5. *Assuming differentiability, we have at $\mathbf{x} = \mathbf{x}(\boldsymbol{\omega}, y)$ that*

- *$\partial c^{SR}(\boldsymbol{\omega}, y, \mathbf{x}(\boldsymbol{\omega}, y))/\partial\overline{x}_j = 0$.*
- *$c^{SR}(\boldsymbol{\omega}, y, \mathbf{x}(\boldsymbol{\omega}, y))$ and $c(\boldsymbol{\omega}, y)$ are tangent at y.*
- *$\mathrm{AC}^{SR}(\boldsymbol{\omega}, y, \mathbf{x}(\boldsymbol{\omega}, y))$ and $\mathrm{AC}(\boldsymbol{\omega}, y)$ are tangent at y.*
- *$\mathrm{MC}^{SR}(\boldsymbol{\omega}, y, \mathbf{x}(\boldsymbol{\omega}, y)) = \mathrm{MC}(\boldsymbol{\omega}, y)$.*

Remark 5.6. It is not necessarily true that $\mathrm{MC}^{\mathrm{SR}} \ge \mathrm{MC}$.

6. Profit Maximization

The FOC of the one-step problem gives $p\partial f(\mathbf{x})/\partial x_i = \omega_i$ for each i, where $p\partial f(\mathbf{x})/\partial x_i$ is called the **marginal revenue product** of input i, or MRP_i.

Proposition 6.1. *Strict concavity of production functions guarantees that the solution to the profit-maximization problem, if it exists, is unique.*

Proposition 6.2 (Output Supply, Input Demand)**.**

- *The output supply function $y(p, \boldsymbol{\omega})$ and the input demand function $\mathbf{x}(p, \boldsymbol{\omega})$ are homogeneous of degree zero.*
- *The substitution matrix*

$$\begin{bmatrix} \frac{\partial y}{\partial p} & \frac{\partial y}{\partial\omega_1} & \cdots & \frac{\partial y}{\partial\omega_n} \\ -\frac{\partial x_1}{\partial p} & -\frac{\partial x_1}{\partial\omega_1} & \cdots & -\frac{\partial x_1}{\partial\omega_n} \\ \vdots & \vdots & \ddots & \vdots \\ -\frac{\partial x_n}{\partial p} & -\frac{\partial x_n}{\partial\omega_1} & \cdots & -\frac{\partial x_n}{\partial\omega_n} \end{bmatrix}$$

is symmetric and positive semidefinite. In particular, $\partial y/\partial p \ge 0$ and $\partial x_i/\partial\omega_i \le 0$. The input demand cannot be inferior.

Proposition 6.3 (Profit Funtion)**.** *If f is continuous, strictly increasing, and strictly quasi-concave, then for all $p \ge 0$ and $\boldsymbol{\omega} \gg 0$, the profit function $\pi(p, \boldsymbol{\omega})$, if well-defined, is:*

- *Continuous.*
- *Increasing in p.*
- *Deceasing in $\boldsymbol{\omega}$,*

- *Convex and homogeneous of degree one in* (p, ω).
- *Differentiable in* $(p, \omega) \gg 0$.
- *(Hotelling's Lemma):* $\partial\pi(p,\omega)/\partial p = y(p,\omega)$; $\partial\pi(p,\omega)/\partial\omega_i = -x_i^*(p,\omega)$.

Proposition 6.4. *From the identity* $\mathbf{x}(p,\omega) = \mathbf{x}(\omega, y(p,\omega))$, *we obtain a Slutsky equation for the input demand function:*

$$\begin{aligned}\frac{\partial x_i(p,\omega)}{\partial \omega_j} &= \frac{\partial x_i(\omega,y)}{\partial \omega_j} + \frac{\partial x_i(\omega,y)}{\partial y}\cdot\frac{\partial y(p,\omega)}{\partial \omega_j}\\ &= \frac{\partial x_i(\omega,y)}{\partial \omega_j} - \frac{\partial x_i(\omega,y)}{\partial y}\cdot x_j(p,\omega),\end{aligned}$$

where the second term is called the ***scale effect****.*

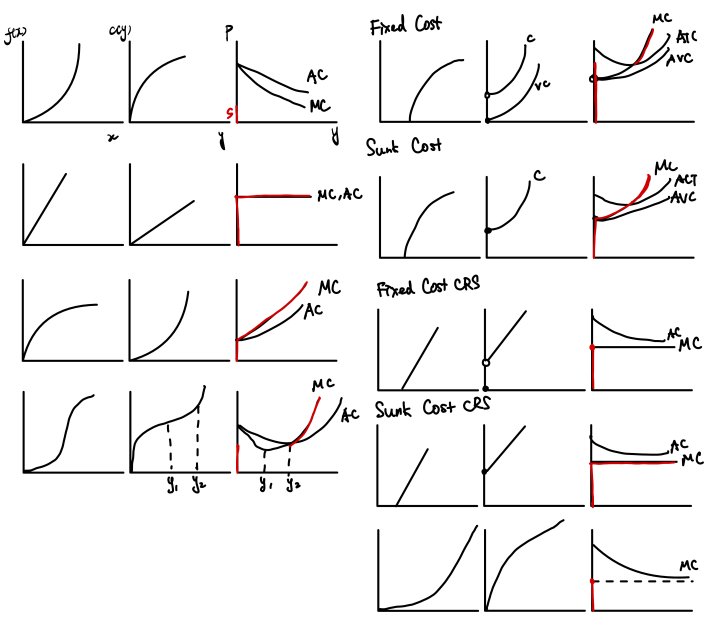
- For $i = j$, recall that $\partial x_i(\omega,y)/\partial\omega_i \le 0$. Thus input demand always satisfies the law of demand and is never Giffen.

6.1. **Marginal and Average Cost.** $\mathrm{MC}(y) := \partial c/\partial y$; $\mathrm{AC}(y) := c/y$. Note that we have

$$\frac{\mathrm{d}\,\mathrm{AC}(y)}{\mathrm{d}y} = \frac{y\cdot\mathrm{MC}(y) - c}{y^2} = \frac{1}{y}(\mathrm{MC}(y) - \mathrm{AC}(y)).$$

Thus the average cost is increasing [decreasing] if and only if it is strictly lower [higher] than the marginal cost. Moreover, L'Hopital shows that $\lim_{h\to 0^+}\mathrm{MC}(h) = \mathrm{AC}(h)$.

6.2. **Firm Behavior.** Recall that the FOC of the profit maximization problem gives $p = \mathrm{MC}$ and the SOC gives $\partial^2 c/\partial y^2 \ge 0$, that is MC must be increasing at the optimal y.
If a firm is allowed to shut-down at no cost, that is, if $f(\mathbf{0}) = 0$, then its profit is bounded below by zero, and the firm will only supply a positive quantity if total revenue exceeds total cost, that is, if $p \ge \mathrm{AC}$. We know thus that the (positive part of the) supply curve is exactly portion of the firm's marginal cost curve that (1) is increasing, and (2) lies above the firm's average cost curve. Note also that the supply curve includes part of the p axis when $\mathrm{MC} < \mathrm{AC}$, which corresponds to the occasions when the firm shuts down.



- AC corresponds to the slope of the secant starting from the origin and MC corresponds to the derivative.
- For the CRS case, note that the supply curve is not defined when $p > \mathrm{MC}$, since the producer can always gain more profit by increasing supply.
- For the fixed cost plots, note that in the cost function is always a minimized value, and thus there is a discontinuity at 0.
- Note that we shutdown if and only if $p\cdot 0 - c(0) \ge py^* - c(y^*)$.
  - Recoverable fixed cost: $c(0) = 0$. Thus shutdown if $py^* - [\kappa + vc(y^*)] \le 0$, or equivalently, if $p \le \mathrm{AC}(y^*)$.
  - Sunk fixed cost: $c(0) = \kappa$. Thus shutdown if $py^* - [\kappa + vc(y^*)] \le -\kappa$, or equivalently, if $p \le \mathrm{AVC}(y^*)$.
  - In the long run, marginal cost equals average cost and is at the minimum average cost.

7. Partial Equilibrium

Definition 7.1. Fixing prices in other markets $(\mathbf{p}, \omega)$, and consumers' income levels m^i, $i = 1, \dots, I$, the **competitive partial equilibrium** is the price p^* such that

- Each consumer i maximizes her utility.
- Each producer j maximizes her profit.
- (Market clearing) The market clears $q^d(p^*) = q^s(p^*)$.

For long-run equilibrium, we impose also a zero-profit condition.
The **compensating variation** is the amount of income that satisfies

$$v^i(\mathbf{p}^1, m^i + CV) = v^i(\mathbf{p}^0, m^i).$$

We may also write

$$\begin{aligned}CV = e(\mathbf{p}^1, u^0) - e(\mathbf{p}^0, u^0) &= \int_{p_j^o}^{p_j^1} \frac{\partial}{\partial p_j} e(\mathbf{p}, u^0)\,\mathrm{d}p_j\\ &= \int_{p_j^o}^{p_j^1} x_j^h(\mathbf{p}, u^0)\,\mathrm{d}p_j.\end{aligned}$$

We similarly define the **consumer surplus** as

$$|\Delta CS| = \int_{p_j^o}^{p_j^1} x_j^m(\mathbf{p}, u^0)\,\mathrm{d}p_j.$$

8. General Equilibrium

Fix exchange economy $\mathcal{E} := (u^i, \mathbf{e}^i)_i$.

- The **contract curve** is the set of allocations where consumers' indifference curves are tangent.
- The **core** of $\mathcal{E}$, $C(\mathcal{E})$, is the set of feasible allocations that are not blocked by any coalition.
- $z_k(\mathbf{p}) := \sum x_k^i - \sum e_k^i$ is the **aggregate excess demand** for good k.
- $\mathbf{p}^* \in \mathbb{R}_+^n$ is a **Walrasian equilibrium** if $\mathbf{z}(\mathbf{p}^*) = \mathbf{0}$.
- $X^W(\mathbf{p}^*(\mathcal{E}), \mathcal{E})$ is the set of equilibrium allocations at price $\mathbf{p}^*$.
- $W(\mathcal{E}) := X^W \cup_{\mathbf{p}^*} (\mathbf{p}^*(\mathcal{E}), \mathcal{E})$ is the set of all equilibrium allocations.

Definition 8.1. Let any nonempty subset $S \subset \mathcal{I}$ of consumers be a **coalition**. We say S **blocks** $\mathbf{x} \in F(\mathbf{e})$ if there is an allocation $\mathbf{x}'$ such that $\sum_{i\in S}(\mathbf{x}')^i = \sum_{i\in S}\mathbf{e}^i$ and $\mathbf{x}'$ is a Pareto improvement to $\mathbf{x}$ for S.
Assumption 8.2. For each consumer $i \in \mathcal{I}$, $u^i : \mathbb{R}_+^n \to \mathbb{R}$ is continuous, strictly quasiconcave, and strongly increasing.
Theorem 8.3. *Under Assumption 8.2, for each* $\mathbf{p} \gg 0$, *the consumer's utility maximization problem has a unique solution. Moreover,* $\mathbf{x}$ *is continuous in* $\mathbf{p}$ *at all* $\mathbf{p} \gg 0$.
Theorem 8.4. *Under Assumption 8.2, for each* $\mathbf{p} \gg 0$, *we have*

- *Continuity:* $\mathbf{z}$ *is continuous at* $\mathbf{p}$;
- *Homogeneity:* $\mathbf{z}(\lambda\mathbf{p}) = \mathbf{z}(\mathbf{p})$ *for all* $\lambda > 0$.
- *Walras' Law:* $\mathbf{p}\cdot\mathbf{z}(\mathbf{p}) = 0$.
- *Existence: if* $\sum \mathbf{e}^i \gg 0$ *then there exists a* $\mathbf{p}^* \gg 0$ *such that* $\mathbf{z}(\mathbf{p}^*) = 0$.

Theorem 8.5 (First Welfare Theorem)**.** *If each* u^i *is strictly increasing, then* $W(\mathcal{E}) \subset C(\mathcal{E})$. *WEAs are Pareto efficient.*
Theorem 8.6 (Second Welfare Theorem)**.** *Suppose Assumption 8.2 holds and* $\sum \mathbf{e}^i \gg \mathbf{0}$. *If* $\overline{\mathbf{x}}$ *is a Pareto efficient allocation of* $\mathcal{E}$, *then it is a Walrasian equilibrium allocation with some price* $\overline{\mathbf{p}}$ *of an economy* $\overline{\mathcal{E}} := (u^i, \overline{\mathbf{e}}^i)_{i\in\mathcal{I}}$ *such that* $\overline{\mathbf{e}}$ *is a feasible allocation in* $\mathcal{E}$ *and* $\overline{\mathbf{p}}\cdot\overline{\mathbf{e}}^i = \overline{\mathbf{p}}\cdot\overline{\mathbf{x}}^i$ *for each* $i \in \mathcal{I}$.

9. General Equilibrium of Production Economy

The economy: $(u_i, \mathbf{e}^i, \theta_{ij}, Y_j)_{i\in\mathcal{I}, j\in\mathcal{J}}$.

- Firm j's production possibility set is $Y^j \subset \mathbb{R}^n$. Its profit maximization problem is $\max_{\mathbf{y}^j \in Y^j} \mathbf{p}\cdot\mathbf{y}^j$.
- The **aggregate production possibilities** is $Y := \sum Y^j$.
- i owns θ_{ij} of firm j. $\sum_i \theta_{ij} = 1$. $m_i(\mathbf{p}) = \mathbf{p}\cdot\mathbf{e}^i + \sum_j \epsilon^{ij}\pi^j(\mathbf{p})$.
- The **aggregate excess demand function** for good k: $z_k(\mathbf{p}) := \sum_i x_k^i(\mathbf{p}, m^i(\mathbf{p})) - \sum_j y_k^j(\mathbf{p}) - \sum \mathbf{e}_k^i$.
- An allocation $(\mathbf{x}, \mathbf{y})$ is **feasible** if $\mathbf{x}^i \in \mathbb{R}^n_+$, $\mathbf{y}^j \in Y^j$ and $\sum \mathbf{x}^i = \sum \mathbf{e}^i + \sum \mathbf{y}^j$.

Assumption 9.1. $\mathbf{0} \in Y^j \subset \mathbb{R}^n$. Y^j is compact and strongly convex.

Proposition 9.2. *If Y^j satisfies Assumption 9.1, then so does Y.*

Theorem 9.3. *Under Assumption 9.1, for each $\mathbf{p} \gg 0$, the solution to the firm's profit maximization problem exists and is unique. Moreover, its supply function $\mathbf{y}^j(\mathbf{p})$ is continuous on $\mathbb{R}^n_{++}$ and its profit function $\pi^j(\mathbf{p})$ is continuous on $\mathbb{R}^n_+$.*

Theorem 9.4. *If each Y^j satisfies Assumption 9.1, then $m^i(\mathbf{p})$ is continuous on $\mathbb{R}^n_+$. In addition, if u^i satisfies Assumption 8.2, then for all $\mathbf{p} \gg 0$, a solution to consumer i's utility maximization problem exists and is unique. Moreover, the Marshallian demand $\mathbf{x}^i(\mathbf{p}, m^i(\mathbf{p}))$ is continuous in $\mathbf{p}$ on $\mathbb{R}^n_{++}$.*

Total profit is maximized if and only if profit is maximized locally:

Theorem 9.5. *For any price $\mathbf{p} \geq 0$, we have $\overline{\mathbf{y}} \in \arg\max_{\mathbf{y}\in Y} \mathbf{p}\cdot\mathbf{y}$ if and only if there exists $(\overline{\mathbf{y}}^j)_j$ such that $\overline{\mathbf{y}}^j \in Y^j$ for all $j \in \mathcal{J}$, $\sum \overline{\mathbf{y}}^j = \overline{\mathbf{y}}$ and $\overline{\mathbf{y}}^j \in \arg\max_{\overline{\mathbf{y}}^j \in Y^j} \mathbf{p}\cdot\mathbf{y}^j$.*

Theorem 9.6 (Existence of GE)**.** *If each u^i satisfies Assumption 8.2, each Y^j Assumption 9.1, and there exists a $\mathbf{y} \in Y$ such that $\sum \mathbf{e}^i + \mathbf{y} \gg 0$, then there exists at least one $\mathbf{p}^* \gg 0$, such that $\mathbf{z}(\mathbf{p}^*) = \mathbf{0}$.*

Theorem 9.7 (First Welfare Theorem, Production Economy)**.** *If each u^i is strictly increasing and $(\mathbf{x}(\mathbf{p}^*), \mathbf{y}(\mathbf{p}^*))$ is a Walrasian equilibrium allocation, then it is Pareto efficient.*

Theorem 9.8 (Second Welfare Theorem, Production Economy)**.** *Suppose each u^i satisfies Assumption 8.2, each Y_j Assumption 9.1, and $\mathbf{y} + \sum \mathbf{e}^i \gg 0$ for some $\mathbf{y} \in Y$. Then for each Pareto efficient allocation $(\hat{\mathbf{x}}, \hat{\mathbf{y}})$, there exists income transfer $T_1, \ldots, T_I \in \mathbb{R}$ satisfying $\sum T_i = 0$ and a price vector $\overline{\mathbf{p}}$ such that*

- *$\hat{\mathbf{x}}_i$ maximizes $u^i(\mathbf{x}^i)$ subject to $\overline{\mathbf{p}}\cdot\mathbf{x}^i \leq m^i(\overline{\mathbf{p}}) + T_i$ for all i, and*
- *$\overline{\mathbf{y}}^j$ maximizes $\overline{\mathbf{p}}\cdot\mathbf{y}^j$ subject to $\mathbf{y}^j \in Y^j$ for all j.*

Thus $(\hat{\mathbf{x}}, \hat{\mathbf{y}})$ is the WEA for the WE $\overline{\mathbf{p}}$.

In general we have

$$\mathrm{WEA}(\mathcal{E}) \subseteq C(\mathcal{E}) \subseteq \mathrm{POA}(\mathcal{E}), \quad \mathrm{CC}(\mathcal{E}) \subseteq \mathrm{POA}(\mathcal{E}).$$

10. Strategic Games

Definition 10.1. An action profile $a^* \in A$ is a **Nash equilibrium** if for each $i \in I$, $u_i(a_i^*, a_{-1}^*) \geq u_i(a_i, a_{-i}^*)$ for all $a_i \in A_i$. That is, if there is no **profitable unilateral deviation** for any player i.

Proposition 10.2. *a a Nash equilibrium if and only if a_i is a best response to a_{-i} for each i.*

11. Monopoly

The cost function: $C(q) = cq$. The monopolist solves

$$\max_q R(q) - C(q) = P(q)\cdot q - c\cdot q.$$

The FOC is

$$R'(q_m) = P(q_m) + P'(q_m)q_m = c.$$

Since $P' < 0$, R' lies below P. From $P(q_{\text{comp}}) = c$ we know that $q_m < q_{\text{comp}}$ and $p_m > p_{\text{comp}}$. We can rewrite the FOC as

$$\begin{aligned} c &= p_m + \left.\frac{\mathrm{d}p}{\mathrm{d}q}\right|_{q_m} \cdot \frac{q_m}{p_m}\cdot p_m \\ &= p_m + \left(1 + \left.\frac{\mathrm{d}p}{\mathrm{d}q}\right|_{q_m} \cdot \frac{q_m}{p_m}\cdot p_m\right) = p_m\left(1 - \frac{1}{|\epsilon(q_m)|}\right), \end{aligned}$$

where $\epsilon(q)$ it the own-price elasticity at q. Thus:

Proposition 11.1.

$$p_m = \frac{c}{1 - \frac{1}{\epsilon(q_m)}} = \frac{|\epsilon(q_m)|}{|\epsilon(q_m)| - 1}c.$$

Remark 11.2. The monopolist always operates at the elastic portion of the demand curve.

12. Cournot Oligopoly

- Homogeneous good.
- Cost functions: $C_i(q_i) = cq_i$ for firms $i = 1, 2$.
- Firm chooses output level.
- Inverse demand function:

$$P(Q) := \begin{cases} \alpha - Q, & Q \leq \alpha, \\ 0, & Q > \alpha. \end{cases}$$

- Payoff functions: $\pi_i(q_1, q_2) := q_i(P(q_1 + q_2) - c)$.

Definition 12.1. (q_1^*, q_2^*) is a **(Cournot-)Nash equilibrium** if q_1^* maximizes firm 1's profit taking q_2^* as given, and vice versa.

13. Bertrand Oligopoly

- Same inverse demand function as Cournot.
- Payoff functions:

$$\pi_i(p_1, p_2) = \begin{cases} (p_i - c)(\alpha - p_i), & p_i < p_j, \\ \frac{1}{2}(p_i - c)(\alpha - p_i), & p_i = p_j, \\ 0, & p_i > p_j. \end{cases}$$

Proposition 13.1. *The unique Nash equilibrium is $p_1^* = p_2^* = c$.*

Appendix A: Production

Cobb-Douglas.

- Production function: $f(\mathbf{x}) = Ax_1^{\alpha}x_2^{1-\alpha}$.
- Cost minimization problem:

$$x_1(\boldsymbol{\omega}, y) = \frac{y}{A}\left(\frac{\omega_2}{\omega_1}\frac{\alpha}{1-\alpha}\right)^{1-\alpha}, \quad x_2(\boldsymbol{\omega}, y) = \frac{y}{A}\left(\frac{\omega_1}{\omega_2}\frac{1-\alpha}{\alpha}\right)^{\alpha}$$

$$c(\boldsymbol{\omega}, y) = \frac{\omega_1^{\alpha}\omega_2^{1-\alpha}}{\alpha^{\alpha}(1-\alpha)^{1-\alpha}}\frac{y}{A}.$$

Cobb-Douglas Generalized, Two Goods.

- Production function: $f(\mathbf{x}) = Ax_1^{\alpha}x_2^{\beta}$.
- Cost minimization problem:

$$x_1(\boldsymbol{\omega}, y) = \left(\frac{y}{A}\right)^{\frac{1}{\alpha+\beta}}\left(\frac{\alpha}{\beta}\frac{\omega_2}{\omega_1}\right)^{\frac{\beta}{\alpha+\beta}}, \quad x_2(\boldsymbol{\omega}, y) = \left(\frac{y}{A}\right)^{\frac{1}{\alpha+\beta}}\left(\frac{\beta}{\alpha}\frac{\omega_1}{\omega_2}\right)^{\frac{\alpha}{\alpha+\beta}}$$

$$c(\boldsymbol{\omega}, y) = (\alpha+\beta)\frac{\omega_1^{\frac{\alpha}{\alpha+\beta}}\omega_2^{\frac{\beta}{\alpha+\beta}}}{\alpha^{\frac{\alpha}{\alpha+\beta}}\beta^{\frac{\beta}{\alpha+\beta}}}\left(\frac{y}{A}\right)^{\frac{1}{\alpha+\beta}}.$$

- Profit Maximization: when $\alpha+\beta < 1$,

$$\pi = (1-\alpha-\beta)\left(\frac{\alpha^{\alpha}\beta^{\beta}}{\omega_1^{\alpha}\omega_2^{\beta}}Ap\right)^{\frac{1}{1-\alpha-\beta}}, \quad y = \left(\frac{\alpha^{\alpha}\beta^{\beta}}{\omega_1^{\alpha}\omega_2^{\beta}}A\right)^{\frac{1}{1-\alpha-\beta}} p^{\frac{\alpha+\beta}{1-\alpha-\beta}},$$

$$x_1 = \left(\frac{\alpha^{\beta-1}\beta^{\beta}}{\omega_1^{\beta-1}\beta^{\beta}}Ap\right)^{\frac{1}{1-\alpha-\beta}}, \quad x_2 = \left(\frac{\alpha^{\alpha}\beta^{\alpha-1}}{\omega_1^{\alpha}\beta^{\alpha-1}}Ap\right)^{\frac{1}{1-\alpha-\beta}}.$$

Cobb-Douglas Generalized, Multiple Goods.

- Production function: $f(\mathbf{x}) = A\prod x_i^{\alpha_i}$. Define

$$\alpha \coloneqq \sum \alpha_i, \quad A \coloneqq \prod \alpha_i^{\alpha_i}, \quad W \coloneqq \prod \omega_i^{\alpha_i}.$$

- Cost minimization problem:

$$x_i(\boldsymbol{\omega}, y) = \frac{a_i}{\omega_i}(W/A)^{1/\alpha}\, y^{1/\alpha}, \quad c(\boldsymbol{\omega}, y) = \alpha\,(W/A)^{1/\alpha}\, y^{1/\alpha}.$$

- Profit maximization: when $a < 1$,

$$\pi = (1-\alpha)p^{\frac{1}{1-\alpha}}(W/A)^{-\frac{1}{1-\alpha}},$$

$$y = p^{\frac{\alpha}{1-\alpha}}(W/A)^{-\frac{1}{1-\alpha}}, \quad x_i = \frac{\alpha_i}{\omega_i}p^{\frac{1}{1-\alpha}}(W/A)^{-\frac{1}{1-\alpha}}.$$

Leontief, Perfect Complements.

- Production function:

$$f(\mathbf{x}) \coloneqq \min\{\alpha x_1, \beta x_2\}.$$

- Cost minimization:

$$x_1 = y/\alpha, \quad x_2 = y/\beta, \quad c(\boldsymbol{\omega}, y) = \left(\frac{\omega_1}{\alpha} + \frac{\omega_2}{\beta}\right)y.$$

Linear, Perfect Substitutes.

- Production function: $f(\mathbf{x}) = \alpha x_1 + \beta x_2$.
- Cost minimization:

$$c(\boldsymbol{\omega}, y) = \min\left\{\frac{\omega_1}{\alpha}, \frac{\omega_2}{\beta}\right\}y, \quad \mathbf{x} = \begin{cases} \left(\frac{y}{\alpha}, 0\right), & \frac{\omega_1}{\alpha} < \frac{\omega_2}{\beta} \\ \left(0, \frac{y}{\beta}\right), & \frac{\omega_1}{\alpha} > \frac{\omega_2}{\beta} \\ \text{convex combo}, & \text{o.w.} \end{cases}$$

- Profit maximization: if $c \coloneqq \omega_1/\alpha < \omega_2/\beta$:

$$\pi = \begin{cases} 0, & p \le c \\ \infty, & p = c \end{cases}, \quad y = \begin{cases} 0, & p < c \\ \mathbb{R}_+, & p = c \\ \infty, & p = c \end{cases}, \quad x_1 = \frac{y}{\alpha}.$$

  If $c \coloneqq \omega_1/\alpha = \omega_2/\beta$: same π, y;

$$\mathbf{x} = \begin{cases} \mathbf{0}, & p < c \\ \mathbb{R}_+^2, & p = c \\ x_1 \text{ or } x_2 = \infty, & p = c \end{cases},$$

Appendix B: Two consumers, Edgeworth Box

$u^i(\mathbf{x}^i) = (x_1^i)^{\alpha}(x_2^i)^{\beta}$

- $x_1^i/x_2^i = \sum e_1/\sum e_2$.
- The contract curve in the Edgeworth box is the line segment connecting the two origins.

$u^1 = (x_1^1)^{\alpha}(x_2^1)^{1-\alpha}$ and $u^2 = (x_1^2)^{\beta}(x_2^2)^{1-\beta}$ with $1 > \alpha > \beta > 0$.

- $y_2^1 = \frac{\frac{\beta}{1-\beta}Yx_1^1}{\frac{\alpha}{1-\alpha}X + \left(\frac{\beta}{1-\beta} - \frac{\alpha}{1-\alpha}\right)x_1^1}$.